Declarations

Conformal Surprisal Fusion for Detecting Prompt-Injection Compromise in LLM Agent Traces

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Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Data availability

All code, the collected agent trajectory corpora, the cached model-judge scores and the analysis scripts that regenerate every number, table and figure reported in this paper are publicly available in the project's GitHub repository.[1] Because the trajectories and the judge scores are released together with the code, the complete evaluation reproduces offline, without access to any model API.

Ethics statement

This work studies defences against indirect prompt injection in LLM agents. All attacks are executed against sandboxed benchmark environments (AgentDojo and τ-bench) using their own published attack templates; no live system, third-party service, or human subject was targeted or involved at any point. No personal data was collected. The attack templates used are already public, so the paper introduces no new offensive capability; its contribution is defensive, and it additionally reports the limits of the proposed defence so that practitioners do not over-rely on it.

Declaration of generative AI use in the writing process

During the preparation of this work the author used a large language model as a coding and editing assistant, for implementation support, for text editing, and as the model-judge component that is itself an object of study in the method (Section 6.1, where the judge model is named). The author reviewed and edited all content, verified every reported number against the released code and data, and takes full responsibility for the content of the publication.

[1] `https://github.com/vinhqdang/SENTRY-Anytime-Valid-E-Process-Monitoring-of-LLM-Agent-Action-Streams`